Mohamed Amin Makni

Software Engineer | AI, Computer Vision & Full-Stack Development

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Profile

Software engineering graduate interested in software development and artificial intelligence. I enjoy solving problems, understanding how systems work and learning new skills, and I'm looking to grow within a collaborative engineering team.

Education

National Engineering Degree in Computer Science | *IIT Sfax* | *Tunisia* **2023 – 2026**

- Specialization: Software Engineering & Business Intelligence. Graduated July 2026.

Preparatory Classes for Engineering Schools | *IPEIS Sfax* | *Tunisia* **2021 – 2023**

Experience

AI Engineer Intern (Final-Year Project) | *Essilor SIVO (EssilorLuxottica)* | *Sfax, Tunisia* **Feb 2026 – Jul 2026**

- Developed a document-processing pipeline and a mobile app for opticians to photograph forms, verify data, submit orders and track status.
- Trained a form-field detector achieving **95.16% mAP@50** and a classifier to distinguish empty and completed fields.
- Gathered business requirements, prepared training data, evaluated models and integrated the pipeline with the backend.
- Achieved **92.42% field-level accuracy** on 54 evaluated order forms, including **97.38% on prescription fields**.

  Tools: *Python, React Native, PyTorch, YOLO11, EfficientNetV2-S, TrOCR, GLM-OCR, RapidFuzz*

Backend Developer Intern | *Altavo Partners* | *Remote (Paris, France)* **Jul 2025 – Aug 2025**

- Built the banking and arbitration module of Hex-Port, a trade-finance platform for African exporters: KYC review, document validation, disputes and bank authentication.

  Tools: *TypeScript, Node.js, Express, PostgreSQL, Prisma, Hedera SDK, Solidity*

Full-Stack Developer Intern | *Infotech Consulting Services (ICS)* | *Sfax, Tunisia* **Jul 2024 – Aug 2024**

- Developed a web application to manage university exams and teaching assignments: sessions, invigilation, teaching loads, timetables and registrations.

  Tools: *Laravel, MySQL, Vue.js*

Projects

ORYS **2026 – Present**

- Develop the AI assistant infrastructure for a business management SaaS.
- Implement provider adapters, validate tool calls, track token usage and costs, and test responses and execution.

  Tools: *TypeScript, OpenAI & Anthropic SDKs, Zod, Vitest*

Fytrak (solo) **2026**

- Independently built the mobile app, admin console and backend of a platform connecting coaches and trainees.
- Linked training and nutrition plans to activity logs and progress, with daily reports and in-app messaging.

  Tools: *React Native (Expo), Firebase, Cloud Functions, TypeScript*

SmartWarehouse AI (academic project) **2025**

- Built an intelligent warehouse-management system combining computer vision, LLM reasoning and RAG to identify arriving vehicles, match them to orders and recommend loading docks.
- Trained a detector for Tunisian licence plates and connected plate recognition to order records and warehouse rules.

  Tools: *Python, FastAPI, Next.js, YOLO, Gemini OCR, ChromaDB, SQL, Docker*

Technical Skills

Programming Languages: Python, Java, TypeScript, JavaScript, SQL, PHP
AI & ML: PyTorch, TensorFlow, scikit-learn, YOLO, OCR (TrOCR, GLM-OCR), RAG, LLM agents (OpenAI, Anthropic), ChromaDB
Backend: Spring Boot, FastAPI, Node.js, Express, Prisma, Laravel, REST APIs
Frontend & Mobile: React, Next.js, React Native (Expo), Vue.js
Data: PostgreSQL, MySQL, MongoDB, Firestore, pandas
Cloud & DevOps: Docker, Git, GitLab CI, Linux, Google Cloud (Firebase, Cloud Functions)

Certifications & Additional Information

Certifications: Information Technology Specialist (ITS) – Artificial Intelligence (Certiport, 2024, 889/1000); ITS – Python (Certiport, 2024, 880/1000); Hashgraph Developer (The Hashgraph Association, 2025)
Languages: Arabic, French, English and some German
Other: SIVP-eligible (Tunisia); open to relocation